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Guozhen Yang

# Compass on the Ocean: Chinese Maritime Theory, History and Culture

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FOREIGN LANGUAGE TEACHING AND RESEARCH PRESS

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# Translators' Words

This monograph titled *Compass on the Ocean: Chinese Maritime Theory, History and Culture* is authored by Prof. Guozhen Yang of the Department of History, Xiamen University. The monograph is a theoretical study of China's maritime development and maritime humanities and social sciences, as well as a study of China's maritime historiography. Maritime historiography, according to the author, sets its sights on the historical studies of all aspects of nature, society, and humanities in relation to the sea, and each of the relevant concepts within the scope of maritime history is elucidated therein. Professor Yang, in particular, gives a precise definition of some of the basic concepts of maritime humanities and social sciences as well as the humanistic studies in maritime sphere including maritime economies, maritime society, and maritime culture, etc., hence clearly establishing the theoretical foundation for the discipline of maritime humanities and social sciences.

The important points that Prof. Yang puts forward about China's maritime history in the book include: (1) China is a large country with both land and sea, rich in bio-diversity, and the ocean is an important environment for the Chinese nation to survive and thrive. (2) In history, the Chinese nation has developed its own maritime economy, maritime society, and maritime humanistic model. As a natural body, the ocean treats all littoral nations equally, and it is neither open nor closed to any one of them. (3) Due to the different marine environments, the ways the coastal nations adapt vary accordingly, hence maritime culture is characterized by plurality and diversity. (4) The contacts and exchanges between maritime civilizations in history are riddled with misunderstandings, exclusions, conflicts, and confrontations. Therefore, it is necessary to develop dialogues among varied maritime civilizations of varied countries, which, as a result, will develop together through mutual tolerance, seek common ground while shelving differences, and extract the idea of maritime civilization for all mankind through mutual learning and complementing each other.

The book is of great academic value: It unveils innovative expositions on the concepts and theories of maritime development, the larger marine science system, and the construction of maritime humanities and social sciences. By utilizing novel ideas and materials and through the historical and global lens, Guozhen Yang has carved out a new realm of comprehensive research in China's maritime humanities and social

sciences, hence providing cultural underpinning for the national strategy to build a stronger maritime country. This masterpiece of his shall become the foundation for the humanistic maritime research and map out the course of its development in the days to come.

The translation of this book is of great significance. The seminal work represents the highest standard of China's domestic scholarship in maritime studies, and also takes a leading position in the arena of the international maritime historiography studies. The publication of the English translation will help to make the Chinese voice heard in the international academic community on maritime issues, increase the international renown of Chinese maritime research, and strengthen China's discourse right in the global academia.

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# Chapter 1
# New Outlook for the Ocean and Maritime Humanities and Social Sciences

## 1.1 Harmony Between Man and the Ocean: New Outlook for the Ocean and Social Development in the Twenty-First Century

Peace and development remain the two themes of top concern in the twenty-first century, but numerous inextricable disputes and conflicts render the world we live in far from being an ideal global village. The kaleidoscopic and panoramic world of today has aroused concerns and reflections from multiple angles. One of the hot issues is maritime development. Maritime development effects social changes through human being's direct or indirect exploration and exploitation of the ocean. Throughout history, human civilizations have emerged locally, giving rise to different styles of civilization in different regions, nations, and countries. Maritime culture, parallel to land-based culture, refers to different patterns of individual and collective ocean-related practices in the process of maritime development. As a natural entity, the ocean is fair to all littorals without discrimination, whether open or closed. Owing to different marine environments, littoral nationalities exhibit different ways of adaptation, hence the plurality and diversity of maritime cultures. Throughout history, the contacts and communications between different civilizations have been riddled with misunderstandings, repulsion, conflicts, and confrontations. Maritime civilizations are no exception. This calls for dialogues across maritime civilizations and borders, common development by fostering mutual tolerance and reserving differences, and shared vision of maritime civilization for mankind through mutual borrowing and learning.

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### *1.1.1 The Expansion of Mankind's Exploration and Exploitation of the Ocean*

On the earth we inhabit, the ocean covers 70.8% of its surface, or an area of 361 million $km^2$. The water it contains is about 14 times that on land. It is the largest natural body on the earth, originally a terra incognita unfit for human habitation. With the invention of boats, ships, and other technical instruments, human beings began to go into the ocean. Their knowledge, exploration, and exploitation of the ocean advanced from the lower to the higher levels, going increasingly deeper from the surface, the body to the multi-dimensional space. From the Challenger's global investigation into the ocean in 1872 to the great discovery of marine resources in the 1960s, the compass of human understanding of the ocean extended from its surface, its body, down to the seafloor and up to the air space. With the advent of the twenty-first century, to match the rise of the strategic position of the ocean, the human society is embracing an era of "multi-dimensional ocean" for full exploration and exploitation of marine resources and space. Our new understanding of the ocean can be concluded in the following four aspects:

(1) The ocean is the second space for human subsistence and development

We used to view the ocean from land-based standpoint, enjoying marine resources as a blessing. The positive significance of the ocean to human subsistence and development seems to consist in its mere role as an artery of communication between civilizations on different continents, connecting littoral peoples as if they were neighbors. As we look back, such a view is obviously one-sided and narrow-minded. Modern marine scientists and humanities and social scientists are well aware that in the maritime regions where human beings live, their economic activities are closely connected with the marine ecological system, which develops into maritime economic ecosystem, and the ocean per se is a space for human subsistence and development.

The ocean is closely linked to human subsistence. It is essential for the survival and development of the species, and is a basic component of life-supporting system. Mankind has migrated from the land to the sea, created navigation-based maritime civilizations, and set up different marine sectors, industrial and social institutions. The ship-dwellers, or "flowing shipboard communities," such as fishermen communities, maritime commerce communities, and pirates, which are themselves space for subsistence and development, have their own organizational systems and behavioral patterns that are distinct from land-based communities. Maritime elements permeate into and have become an indispensable part of their material and spiritual lives. Long voyages at sea challenge the sailors physically, physiologically, and mentally as well. As a matter of fact, where there was no capacity for survival, there would be no on-board communities, let alone crossing or exploring the ocean. In history, maritime activities comprised only part of human activities. In coastal regions, islands, and territorial seas which were unfit for human life and navigation and lacking in maritime activities, there naturally was no maritime culture. Over a long period of time, man enjoyed huge space on land, sufficient for them to survive and thrive; maritime

communities were only in the minority. Hence, it's understandable why the mainstream mentality of the human society has long neglected the value of the ocean as a living space.

As mankind entered the twentieth century, with the rapid growth of world population, the living space on land was getting inadequate, and the ocean has become a coveted living space for mankind. The construction of land reclamation, sea airports, offshore oil and gas platforms, cities on the water, underwater habitats, underwater hotels, underwater tunnels, and underwater factories—all mirror this desire. The shortage of land resources, the deterioration of environment, and the thirst for compensation from the ocean have driven mankind to seize and divide the multidimensional ocean space. On 16th November 1994, *The United Nations Convention on the Law of the Sea* (hereinafter the *UNCLOS*) took effect, which has since changed the conventional pattern of the high seas, so that more than 30% of the ocean (1094 million $km^2$) on the globe now falls under the jurisdiction of coastal States. The "internal waters" (within the baseline of territorial seas), "territorial seas" (not exceeding 12 nautical miles from the baseline at which the breadth of the territorial sea is measured), and its air space, sea beds and subsoil—all enjoy the same full rights as land territories do. Other territorial seas, such as "contiguous zones, exclusive economic zones, straits for international navigation, and archipelagic waters," carry different levels of rights, namely autonomous rights, exclusive rights, territorial rights, or administrative rights, respectively. These five types of waters are different from internal waters, territorial seas, and high seas, yet the coastal and archipelagic states have placed the territorial seas under de facto control irrespective of the "de jure restrictions."

(2) The ocean is the pivot of economic development

The ocean is abundant with biological and mineral resources, which are a valuable treasure for human beings to sustain and promote their livelihood. The capacity of the ocean to feed mankind is approximately 1000 times that of agricultural products around the globe. More than 80% of all proteins human beings need are from sea products. We can cultivate salt-tolerant crops by genetic modification and genetic engineering, develop seawater irrigation agriculture, and grow terrestrial plants in the sea, so as to ease the pressure of agriculture on land. Sea water can be used for industrial and municipal purposes and for extraction of chemical elements, which can be processed into various kinds of chemical products. It can even be desalted for domestic use, which has become a major source of sustainable freshwater. Various kinds of marine energy, including tide energy, wave energy, current energy, thermal energy and salinity gradient energy (or blue energy), are renewable resources dependent on sea water, amounting to a total of over three billion kilowatts. The reserves of marine oil and natural gas are estimated to be 1400 billion tons. The international seabed area, which accounts for 49% of the land surface, contains rich resources of polymetallic nodule, cobalt-rich iron / manganese (ferromanganese) crust, and hydrothermal sulfide. Gas hydrates (usually called flammable ice) are found in the seafloor and permafrost sediments along the continental margins at water depths greater than 300 m. One cubic meter of gas hydrate can release 164 cubic meters of

methane gas and 0.8 cubic meter of water, an estimated total of 20,000 trillion cubic meters. This contains an estimated volume of organic carbon twice that of the known total storage of coal, oil, and gas, which can last 1000 years for human use (Yang et al., 2003).

The ocean is fundamental to human life. Inhabiting, swimming, boating, fishing, and sightseeing are the daily activities of the coastal life. Regions where economy, society and culture are best-developed are mostly concentrated in coastal areas, around 60 km from the coast, which accommodate half of the world's population. It was once estimated that about 3600 people were migrating to the coastal areas on a daily basis. Meanwhile, the ocean is of high ecological value: the regulation and intervention of the atmosphere, the circulation of nutritious salt, waste disposal, control of organisms, living environment, food, raw materials, entertainment and other cultural forms, these are all pivotal to harmonious development.

(3) The ocean is the "inner outer-space" for strategic competition

The ocean remains a security line for all nations. The prominence of energy and economic security, and the application of high and new technology in military have bestowed maritime security and the strategic role of the ocean with new meanings. While the conventional strategy of "controlling the world by controlling the marine channels" is not outdated, the priority has gradually shifted to a multi-dimensional ocean, especially to the still unknown "inner outer-space," the 500-m-deep territorial sea. The coastal zones, bay areas and islands are garrisoned by the longshore defense troops; meanwhile, the naval forces are stationed on the sea, the submarine forces under the sea, and missile bases are built on the ocean floor, through which the basic patterns of future naval deployments can be envisaged. The major strategic nuclear forces have shifted from land to ocean, and with missile nuclear submarines and anti-submarine weapons as the main power, they can be deployed in different seas around the world, capable to carry out surprise attacks from different locations, and mitigate the threat of nuclear strikes on their own missile systems and homeland. The uninhabited islands and reefs in the ocean and underwater are ideal sites for weapons testing and military drills. Marine warfare is not limited to naval battles, but a multi-dimensional one that covers the air space above the ocean, islands, underwater and seabed, with the integrated characteristics of land, sea and air warfare concurrently.

(4) The ocean is an important platform for innovation of science and technology

The ocean is an important site in which scientific discoveries are made. The solution for major scientific issues, such as global warming, climate change, the origin of life and the origin of mankind, rests much with the progress of research on marine science. Research of this kind is called ocean mega-science, including the global ocean observing system, marine drilling programs, hydrothermal process and ecosystem, marine biodiversity, integrated administration of coastal zones, etc. Man's understanding of the ocean is still rather shallow, but its potential as a mega-science, and of economic benefits and utilizability has generated increasing interests and enormous concerns. High and new marine technology and maritime industries are springing up and holding out new hope for the development of human civilization.

The ocean is the way out for future civilization. Exploration, exploitation and protection of the ocean are the bound direction to which the human society moves forward in the twenty-first century.

### *1.1.2 Hidden Worries of the Ocean Century*

The twenty-first century is a new era in which mankind explores the three-dimensional ocean. However, human beings still know little of the ocean. Most of them have no clue about it, and some of them have only a smattering of understanding or even misunderstandings regarding it. While we have been trying to gain insights or to rediscover the ocean, our notion of subsistence and development has not changed. In the current sociocultural and economic competition where profit-maximization is most highly valued, the negative consequences of mankind's contention for living space and plundering of resources have rendered the relationship between man and the ocean, and that between individuals and between countries in maritime activities all the more strenuous, which can be specified as follows:

(1) Enclosure movements are increasing in intensity

Human being's contention for living space in the twentieth century started with developed maritime powers and then came the third-world countries, struggling for territorial seas, so that the enclosure movements have escalated over time. Approximately 80% of the fishing areas with edible fish stocks have been divided into territorial seas under the jurisdiction of States concerned. The total catch of fish from the exclusive economic zones of coastal States amounts to more than 94%. Fishing in other States' specialized zones is very much restricted. This means some fishing vessels are forced to withdraw from the traditional fishing grounds, which will seriously impact the livelihoods of fishermen and the economy of the fishing areas. Of the global explored oil storage in the ocean, 87% lies in the exclusive economic zones of the coastal States. The main international navigation routes also go through the exclusive economic zones of one or more States. Therefore, after the *UNCLOS* took effect, all States involved adopted the attitude of fighting for every inch of sea in the matter of territorial waters. Some even broke the limits imposed by the *UNCLOS*, made various claims to expand their maritime rights and took preemptive action in the disputed waters in an attempt to get a larger share as a fait accompli. Free ocean space has been narrowed down and the living space has shrunk as a result. There have been continuous battles for island autonomy, fishery disputes, energy and control of straits, which are potential causes for war.

(2) Populations are flooding into coastal areas

Populations flooding to coastal areas have become a global trend, starting with developed States and followed by developing States. People agglomerate in the coastal areas and build clusters of big cities. While this symbolizes economic prosperity

and modernity, it does bring about a myriad of problems as well, such as insufficiency of living space, tension between man and land, excessive expansion of cities towards the ocean, and fragility of ecological environment. It even results in shortage of freshwater and leads to water scarcity, excessive pumping of underground water, which in turn could cause subsidence and saltwater intrusion. Demands for sea products grow dramatically, which stimulates overfishing and excessive exploitation of mariculture. The rising volume of industrial and domestic wastes and sewage leads to serious pollution. Urban heat island effect and rain island effect are aggravated, which increases the risk of climate change. Measures of governance fail to keep up with the change, which leads to lower quality of life, insufficient employment, stagnation of economy, social disorder, etc. That's why some scholars point out that the threats to the global maritime regions mainly arise from population concentration in the coastal regions.

(3) Inappropriate or excessive exploitation of the ocean

Inspired by the slogan of "constructing mankind's second homeland," coastal States are jumping on a bandwagon of maritime exploration, making maritime economy a new economic growth point. Some of the sea areas, however, are exploited inappropriately and excessively, always overused but under-protected, thus resulting in conflicts between ecological objectives and economic ones.

Marine fishery resource is a special kind of common property. Owing to the mobility of fish and legal ambiguity of property rights, the so-called common property actually means that it doesn't belong to anyone and that everyone can fish for free. That's why large-scale ship-making and long-term overfishing run rampant despite prohibitions. Overfishing has the most tragic consequences for the ocean, resulting in the extinction of some living resources. Marine biologists warn that such acts, if unchecked, may foreshadow a future decades later where man could fish nothing from the ocean but planktons.

The use of the ocean with disorder and without limits has ruined some marine resources and natural landscape. The scarcity of marine fish varieties has upset the balance of species structure and caused ecological disasters such as marine plague (virosis). Coastal constructions such as building dykes on the coast have altered the direction of the currents and the surroundings, which might barricade the routes of migratory fishes and destroy the ecological system. Sand and gravel are recklessly excavated, vast arrays of mangrove forest and reefs have vanished, and wetland is shrinking sharply, which has largely diminished the capacity of the ocean for regulating the climate, water storage and flood diversion, as well as for defense against windstorm torrents and safeguarding the coast. Bringing in exotic species blindly has crowded out and caused the extinction of many native species. Changing the landscape has come at the cost of aesthetic value.

(4) Marine pollution recurs despite prohibitions

Inhabitants on land take the ocean as a natural garbage pool and dump wastes and pollutants into the sea recklessly. Mankind is now placing intolerable strain on the coastal areas by dumping wastes and pollutants into the ocean, by letting the untreated

or poorly-treated industrial, agricultural and domestic sewage flow unchecked into the ocean. Following the increased exploitation, oil-leakage, the discharge of ballast water and waste water from fishery and offshore field further deteriorate the offshore environment. Severe pollution in some parts of the waters has gone beyond the ocean's capacity of thinning and self-purification. Pollution lowers the quality of sea water and causes oxygen deprivation, which directly intoxicates marine lives or causes damage to their normal physiological and biological functions. It also results in eutrophication, which causes explosive reproduction or dense agglomeration of some planktons, which then develop into red tide and suffocates a vast number of fish and shellfish.

If excessive exploitation of the ocean and aggravation of pollution remain unrestrained, the offshore may become unfit for human life, which is sounding the alarm to mankind. That's why the United Nations took "Wanted! Seas and Oceans—Dead or Alive" as its theme on the 33rd World Environment Day, on 5th June 2004.

(5) Anomalous change of marine environment

Marine disasters induced by the change of marine environment, such as catastrophic earthquakes, tsunamis, rainstorms and slides, continual coast erosion and the rise of sea level, can have great impact on the human society. El Nino has resulted in abnormal temperature rise across a vast range of seas, and as a result, a myriad of fish and seabirds living on fish die. Lately in Peru, warm water intrusions have occurred every few years, causing massive deaths of fish and fish-feeding seabirds. The rise of sea level caused by global climate change has pushed the ecosystem of the coastal regions back towards land, which has direct impact on the biodiversity of coastal areas across the world. Studies show that a rise of one meter in sea level will cause a coastal retreat by 1500 m. If the sea level reaches this level 50 years later, it is forecast that Tuvalu, an island country, will be submerged and might be the first country to suffer the nation-wide exodus. America might end up losing 360,000 $km^2$ of its land bit by bit, and coastal States bordering on the Atlantic, the Mississippi River and the Mexican Bay might be engulfed. Likewise, 120,000 $km^2$ of China's coastal areas might be submerged and 70 million people might have to be migrated inland (Yang et al., 2000).

### *1.1.3 Towards Harmony Between Man and the Ocean*

Harmony between man and the ocean should be an ideal of maritime civilization that the 21st-century mankind pursues. It not only marks a new stage of maritime exploration, the brainchild of the reflection on the marine ecological-economic crisis; it is also a re-creation of the ancient philosophy in the Eastern world that man is an integral part of nature. The idea was put forward in the twentieth century and viewed as of universal value in *Agenda 21* adopted in 1992 at the United Nations Conference on Environment and Development and in the *UNCLOS* that took effect on 16th November 1994. Since then, it has become a guiding principle for sustainable

maritime development, aiming to construct a new international order in the fair share of maritime interests, joint exploration and protection and peaceful exploitation of the ocean.

Harmony between man and the ocean involves the adjustment of relationship between man and the ocean and that between individuals. International communities and scholars have given much thought on this, appealed for the construction of an institutionalized working mechanism out of conscience and morality, and have drawn up alternative norms or plan Bs, which has greatly enriched the connotations of the new maritime outlook.

(1) Fair share of maritime interests

From social perspective, we propose that maritime benefits be equally shared, and that coordination in terms of social development should be maintained between maritime regions and land regions, between coastal States and landlocked States, as well as between interpersonal and intergenerational development. This concerns multiple levels of issues.

At the first level, i.e., in local maritime regions per se, such as a bay or a patch of waters as a unit, there assemble different communities, industries and individuals, who explore and exploit the ocean with different purposes and with overlapping or conflicting interests. For instance, fishery and aquaculture industry may conflict with, or even ruin maritime transportation, salt industry, seashore industrial reserves and seaside tourism, which may restrain the regional social development. We must take into account environmental, economic, social and cultural factors, make equitable allocation of the coastline, wetland, territorial seas and resources, rationalize productivity layout and ensure mutual coordination and complementarity in order to optimize the overall performance of the social system in the marine zones. Meanwhile, the activities of maritime communities with respect to production, circulation, distribution and consumption should not adversely affect other uninvolved communities.

At the second level, maritime regions within a country but under different jurisdictions may have different focal points with respect to exploitation and development strategies, which may also incur disputes and conflicts of maritime interests. Between maritime regions and inland areas, between the central and local governments, there is also the issue regarding the distribution of maritime interests, which calls for the adjustment of relationship on the basis of the principle of fair play, such as the ecological compensations for the mitigation of disparities in wealth in different regions.

At the third level, between coastal and island States, especially those that are geographically adjacent or opposite to each other, there may exist disputes and conflicts on the division of territorial seas and the attached maritime rights. This calls for equity in the delimitation of the territorial seas, continental shelf and exclusive economic zones. There also exist disputes and conflicts on maritime interests between geographically advantaged and disadvantaged States or both, and between coastal or island States and landlocked States, which requires redistribution of the actual benefits on equal basis. In accordance with the *UNCLOS*, both landlocked and

shelf-locked States are entitled to a proportion of gains from the exclusive economic zones within the neighboring littoral states. The landlocked States have the right of access to and from the sea and freedom of transit through the territory of transit countries (The *UNCLOS*, 1982: 68). All States enjoy the freedom of navigation and overflight in the exclusive economic zones of other countries, as well as the rights to lay submarine cables and pipelines, and other legitimate uses of the international seas related to the rights. Coastal and island States can sign agreements or issue permits to other States, and their legal persons or natural persons should be allowed to explore and exploit the marine resources independently or jointly. Other States can pay maintenance fees to obtain permission for fishing in the territorial seas. Landlocked states should also be entitled to a fair share of surplus fish from coastal States' territorial seas (The *UNCLOS*, 1982: 49). When a coastal State's fishing capacity approximates the total amount of the living resources available in the exclusive economic zones, the coastal State and other States concerned shall cooperate in the establishment of equitable arrangements on a bilateral, sub-regional or regional basis to allow for participation of developing landlocked States of the same sub-region or region in the exploitation of the living resources of the exclusive economic zones of coastal States of the sub-region or region, as may be appropriate in the circumstances and on terms satisfactory to all parties. Developed landlocked States and geographically disadvantaged States should be allowed to join in exploiting the living resources in the exclusive economic zone under the jurisdiction of the developed coastal States within that region. In practice, agreements should be reached through negotiations on terms satisfactory to all parties (The *UNCLOS*, 1982: 50).

Exploitation of the high seas and seabed areas should also follow the equity principle and legal norms. The *UNCLOS* stipulates that the high seas are open to all nations. All nations share the rights of navigation, overflight, laying submarine cables and pipelines, construction of artificial islands and other facilities approved by the international law, fishing and carrying out scientific research. Not a single country can effectively claim sovereignty over any part of the high seas. The international seabed areas, including the seabed, the ocean floor and sub-soils other than the continental shelves, are common properties of the whole humankind, under the administration of the International Seabed Authority in the name of the whole world. The economic benefits from exploration and exploitation of the seas and their resources should be shared by all nations across the world, taking special care of the developing States (The *UNCLOS*, 1982: 57). According to Article 82 of the *UNCLOS*, to exploit the non-living resources of the continental shelf 200 nautical miles beyond the baselines from which the breadth of the territorial sea is measured, coastal States should make an annual payment of 1% of the value or volume of production at the site to the International Seabed Authority for the sixth year. The rate shall increase by 1 percent for each subsequent year until the twelfth year and shall remain at 7% thereafter. The International Seabed Authority distributes the revenue among the States parties on the basis of equitable sharing criteria (The *UNCLOS*, 1982: 55), and the board of governors of the International Seabed Authority should also be so composed as to be based on equity.

At the fourth level, maritime benefits should be shared fairly across generations. Individuals of the contemporary world should take into consideration not only the equity of opportunity in terms of maritime development and demands and interests of all parties concerned, i.e., interpersonal equality, but also the future generations' rights of subsistence and maritime interests, namely intergenerational equality. As the contemporaries cannot possibly judge exactly what the future generations' needs will be, it's unlikely that we make a specific list of what to be reserved for our descendants, but we can at least morally set aside the resources and interests that are no less than what the preceding generations have saved for us. Excessive consumption will expect us to pay our descendants the rents for ecological environment, and we shall pass down this tradition across generations.

The order of the twenty-first century maritime world is still dominated by the powers, lacking in equity and justice for fair share of marine resources. It's hardly possible for the vulnerable communities and States to enjoy the benefits of maritime development equally; they are even marginalized. There arises a voice which proposes that the commonly inherited properties of mankind should cover the high seas and all territorial seas that have been divided by different States, so as to make the whole maritime space a common property to the whole humankind. But there is still a long way to go. It's been predicted that the basic pattern of the territorial division of the oceans will not change substantially in the twenty-first century. But with joint efforts, there will likely be coexistence and common prosperity or win-win competition among States with equal capabilities. Both coastal States and island States wish to become major maritime countries, but the right approach should be to dialogue rather than conflict, to deal with disputes and conflicts by means of diplomacy and law, to keep the ocean in peace and to coexist with win-win cooperation, rather than at the expense of other nations' well-being.

(2) Sustainable exploitation of marine resources

From economic perspective, we propose sustainable economic growth on the ground of protecting marine eco-environment. This involves transforming the traditional notion of materialistic consumption and development at the cost of environment, so as to explore and exploit marine resources appropriately, and maintain the virtuous cycle of marine resource ecosystem. On the whole, it is desirable that the contemporaries' needs be satisfied within the capacity of resources and environment without jeopardizing the interests of future generations.

In order to exploit marine resources sustainably, the developed maritime countries have initiated and set up the integrated management system of the ocean and coastal zones. With the support and promotion of the World Bank and the related organizations of the United Nations, many countries have started to pilot and expand it. In 1996, the United Nations Intergovernmental Oceanographic Commission carried out a sampling survey and concluded that the integrated management system is a logical framework, with sufficient flexibility to satisfy most, if not all States' different needs.

The integrated management system of the ocean and coastal zones is a complex issue, which involves the adjustment of system design, development strategies and operating mechanism. On the strength of the ongoing worldwide practice, the idea of

integrating coordinative development between economy and ecology, between land and ocean, and between regions and the globe is getting clearer. Scholars in this field have concluded three key principles for sustainable maritime development: (1) the exploitation rate of the living resources should not exceed their replacement rate; (2) the volume of wastes and sewage discharged into the ocean should not exceed the capacity of the oceanic zones; (3) the population in coastal regions should not exceed what these regions and marine ecology can accommodate.

As to specific measures, coastal States are working hard to enhance their capability of exploiting the ocean with science and technology, and to maximize the efficiency of marine resource exploitation at lower costs. The measures can be listed as follows:

1. Strengthen governmental involvement into maritime development via legislation, planning, investment and administration. In other words, a scientific and appropriate plan for how to exploit the territorial seas should be drawn up and the licensing system for the paid use of territorial seas should be implemented. With the establishment of an integrated decision-making mechanism for resource environment and development, it is likely to prevent the destruction of marine resource environment at the very beginning, to make full use of various maritime industries in shallow or deep waters, each with their respective functions, and to guarantee the highly efficient use of marine resources both temporally and spatially.
2. Adopt justified means of maintenance, weigh the relationship between the volume of reproduction and capture production of the living resources, and come up with novel means of administration on fishing. For instance, fishing seasons and fishing areas can be planned, limits can be put on the size and number of fishing vessels and fishing gear, the varieties and quota of fish that are allowed to catch can be regulated by issuing fishing permits to fishermen. Closure of fishing areas, seasons of closure and fishing moratorium need to be stipulated. In the offshore areas, bottom trawl should be banned, and limits should be put on the variety of nets and the size of screens, so as to prevent excessive fishing and to protect the reproduction and breeding of some species. When other countries' fish varieties and fish schools are involved, it is advisable to establish a mechanism of cooperation among countries concerned and other international organizations.
3. Make scientific and proper use of aqua-cultural area near the beach and the shallow sea, develop an efficient low-polluted and large-scale agrarian model, and promote pollution-free breeding. For example, we can develop technology that may control diseases in marine aquaculture, and set up a series of integrated prevention system ranging from early pathogen diagnosis, segregation and elimination, daily health care of hosts, improvement of their immunity and premunition, to the sanitation and purification of breeding environment, control and treatment of pollution. All these are aimed to set up an integrated prevention and treatment system to guarantee the health of cultured organisms. And we need to explore new ways and technologies for coastal breeding, offshore breeding and even breeding in the high seas.

4. Prevent, alleviate and control the damages of pollution that human activities might inflict on marine environment, and prevent marine ecosystem degradation. Environmental protection should shift from focus on pollution prevention to the holistic management of prevention, monitoring and governance, and to the integrated management of resources and environment. These means of exploring and exploiting marine resources ensure the unified sustainability of maritime economic growth, marine ecology and social benefits, the ultimate goal of which is human subsistence and development.

(3) Harmonious coexistence of man and the ocean

Nature-wise, we propose a new type of relationship between man and nature characterized by harmony, equity and coordination, and the harmonious coexistence of man and the ocean is among the most important.

Human life is closely connected with lives of other species, and numerous factors of human being's living environment are subject to the ocean. Mankind's exploration and exploitation of the ocean can also influence its natural evolution. The connotations of harmony between man and the ocean differ in different periods of history. In the modern relationship between man and the ocean, man dominates, and hence the fundamental conception of harmonious development of man and the ocean refers to mutual adaptation and mutual support between human social system and natural marine system against the backdrop of large-scale maritime exploration. In other words, on the one hand, while exploring and exploiting the ocean, mankind should observe the ecological law of the ocean, satisfy the needs of the human society and protect the value and function of the ocean in life support and environment purification. On the other hand, we should set the ocean's natural evolution back on the right track by self-regulating our demands and behavior to mitigate the damages to natural evolution, tap into our wisdom and motivation to extricate the renewable resources (which are vital to human beings) from the arduous and slow mode of self-evolution, and facilitate the coordinative evolution of man and the ocean.

The current pervasive maritime development model is overridden with the Western knowledge-power ideology. Productivity has far surpassed that of any previous age and disharmony between man and the ocean has become far more severe and complex than any time before. Human history is a process of human beings transforming nature and themselves at the same time. When tracing back to the origin of maritime space and the resource-plundering type of civilization, we must reconsider the human development model. Man used to pride himself as master of nature and take it upon himself to conquer nature, extort resources from nature voraciously and pursue development at the cost of nature. In order to change such a unipolar pattern of development, we must first change our mindset, be kinder to the environment we depend on and put effective protection of eco-environment on a par with economic development. Protection must run in parallel with development, making sure human society adapts to the environment. Only in this way can harmonious development be achieved among man, land and ocean.

Based on the above-mentioned philosophy, countries across the world should share the responsibilities and obligations of protecting the ocean, preventing marine

resources from being destroyed and environment from degradation. As a matter of fact, many countries are already taking actions to varying degrees to build a strong and efficient social administration system as part of the institutional improvement of maritime civilization. For example, coastal States are taking measures to protect marine environment within the territorial seas, including putting a cap on the total volume of sewage released into the ocean and implementing discharge fee, as well as the management systems for coastal constructions, waste dumping, pollution incidents reporting and handling in order to prevent, reduce and control any possible marine pollution. We should treasure the coastal landscape, beach and other tourist attractions, and develop maritime culture industries, such as coastal tourism, marine fishing for pleasure, space for tourism and entertainment, which not only entertain the public, meet our needs for leisure and pleasure, but also open up new career options for fishermen and relieve the pressure of marine environment and resources. We should cherish the health and well-being of marine life, create reserves for aquatic germplasm and genetic resources, juvenile fish resources, aquatic wildlife, marine and coastal ecosystem, and rare or endangered species. We should build up circular economy and green maritime economy to strike a balance between marine environment and maritime development. These measures are all oriented towards harmonious development and coexistence of man and the ocean.

Sharing maritime benefits equally, exploiting marine resources sustainably and coexisting with the ocean harmoniously, all these are of universal value. It builds on modern maritime material culture and provides humanistic guidance on the direction of maritime material culture development, which may have increasing impact on the social development and all-round progress in the twenty-first century. The current situation however is alienated from this value. Battles for maritime benefits are increasing, and a fundamental change of the current development at the cost of environment is yet to occur, but we must not lose heart. This ocean century will ultimately be framed as a century of comprehensive maritime development and of maritime peace and health.

## 1.2 Basic Theoretical Research on Maritime Development

Maritime development is a topic of significant theoretical and practical value globally, strategically, and prospectively. The research objective consists in resolving various conflicts and disputes that have occurred in ocean-related human activities, so as to achieve sustainable development in the endeavor of exploration, exploitation, administration, and protection of the ocean. Maritime development research essentially falls under the rubric of humanities and social sciences, interpenetrating and overlapping with the research on the development of marine science and technology. It's an emerging, integrated, and yet marginalized field of research.

The exploration of the world's oceans and the expansion of maritime affairs have shaped the morphology and trajectory of maritime civilization. Timely summaries on them call for theoretical guidance and criteria. There's no denying that previous

theories of maritime development are basically the extended theories of development on land, which do not fit in with modern maritime exploration and expansion. This section aims to make tentative discussions about the basic theories of maritime development.

### *1.2.1 Reasons for Building Theories of Maritime Development*

Maritime development refers to the behavior of human beings in their direct or indirect exploration and exploitation of maritime activities to influence social transition (Yang, 2005). This is a neutral concept, yet the results that are presented by the facts of maritime development can be positive or negative. If our understanding of the facts stops at the perceptual level, then maritime development naturally would not automatically emerge as a theoretical issue and become visible to researchers unless it is elevated to the heights of rational understanding through theoretical constructs.

As we entered the twenty-first century, the essence and complexity of the ocean have fostered marine science into a systemic science, including all natural sciences. Recently, social sciences are also included in it (Field et al., 2002). The complex maritime issue has posed greater demands not only to the academic world of marine science, but to that of humanities and social sciences as well. Therefore, the building of basic theories of maritime humanities and social sciences is doubtlessly of tremendous theoretical value and practical significance.

The basic theories of maritime development, which refer to the theories arguing for the value of human activities in the exploration and exploitation of the ocean, the route of development and social benefits, as well as their status and role in human civilization, may provide guidance for maritime practice.

The actual distinction between the natural ocean and the humanistic ocean offers an objective basis for the theoretical construct of maritime development. The natural ocean refers to the natural combination of water bodies (including reefs, sub-soils, and the surrounding coastal areas) and the air space, which provides the marine environment for human development. The inner elements of the natural ocean interact closely. Any local interference may evoke a global change of the region and can have tremendous impact on the makeup of the atmosphere and the trend of the global climate. The humanistic ocean refers to the behavior, lifestyle, and means of interaction of human beings that are based on the natural ocean. It is a product of the practice that mankind looks upon the oceans as his historical living space and attempts to define the world from the perspective of the ocean rather than land (Schmitt cited from Lin & Zhou, 2006: 93). The geographical distribution is subject to natural and humanistic conditions. The division of maritime space stipulated by the *UNCLOS*, including the distribution of benefits among coastal States, island States, and landlocked States, is not fully consistent with the natural situation and can be viewed as a hybrid of the natural ocean and the humanistic ocean.

The natural ocean shapes the humanistic ocean and vice versa. The exploitation of marine resources, especially seabed oil and gas, and mineral resources, depends on the progress of marine science and technology, and is simultaneously restricted by maritime politics, economy, society, cultural environment, and current international affairs. The protection of China's maritime benefits and peaceful development of the ocean depends on the enhancement of its comprehensive national strength, the breakthrough in maritime politics and diplomacy, the reinforcement from the theory of maritime development, and maritime humanistic spirit. This calls for theoretical guidance for the comprehensive and macro studies of maritime development.

Academically, the basic theory of maritime development is a branch of development theory, yet it's far from being the case in actual practice. Development theory originated from the West. In the mid-eighteenth century, the word "development" (*developpent* in French or *entwicklung* in German) was used to denote the process of social change. With the rise of the Industrial Revolution and the theory of evolution, the connotation of development has changed to progress of production and material life. After the Second World War, development studies "aimed to explain the complicated operating rules in the process of development" (Li, 2005: 18) emerged. Initially, it was focused on the issue of how developing countries emerge from underdevelopment, and later expanded to the development history of developed countries and the issue of sustainable development of the human society as a whole.

Research on development first emerged in economics, giving rise to development economics, focusing on economic growth. It put emphasis on "studies of how economically disadvantaged countries or agricultural countries can achieve industrialization and modernization, the economic takeoff and growth" (Zhang, 2001: 2), and emphatically examined how the average economic conditions and income distributions among individuals or different countries and regions can affect growth (Ray, 1998).

Later, development studies in sociology and political science followed up, each with their own research scope and research model. Development sociology is "a peripheral discipline at the intersection of development studies and sociology, or a research on development from sociological perspective" (Zhang & Ma, 2001: 21). It mainly explores through social dimensions "the fundamental characteristics of modernization," "the model of development and modernization," and "the progression of modernization" (Wu & Liu, 2002: 16). It advocates research through the approaches of social indicators, historical comparison, and policy (ibid: 395). The politics of development takes political development as its theme and discusses how developing countries achieve their democratic transformations, how to maintain social stability, how to combat political corruption, how to resolve political crisis, how to accomplish political and cultural reformation, etc. Development anthropology approaches the differences in the "social status, role, and income of different ethnic groups," and "undertakes studies from all-round perspectives of ethnic culture" (Chen, 2001: 47). "Development philosophy takes a theoretical and systematic view of development…it is based on knowledge drawn from the development of economy,